CSC186 – Object Oriented Programming
Academic Session October 2025 – February 2026
Lab Assignment 2 - Basic Concepts of Classes

Course Outcomes (CO)	LO1	LO2	LO3
CO1			
CO2	√	√	√
CO3			

1.1 Given the `Program` class that consists of the following data members (attributes):

- Program code (eg: CS110)
- Program description (eg: Diploma in Computer Science)
- Program duration (eg: 3 years)
- Faculty (offered by which faculty, eg: Faculty of Computer & Mathematical Sciences)
- Program Head (name of the person incharge)

Write the `Program` class and the following methods:

a) Default constructor
b) Normal constructor that set all data with values given through the parameter.
c) Copy Constructor
d) Mutator/Setter method
e) Retriever method for each attribute.
f) Printer method using `toString()` to return object information.
g) A processor method to return the program level. The program level canbe determined through the third character of program code as shown in table below:

Program Code	Program Level
CS009	Certificate
CS110	Diploma
CS220	Degree
CS770	Master
CS990	Doctorate

Details of Program Level

Example*: CS**0**09 – third character is 0, which is Certificate Program Level*

Write an application program that will read all attributes and store them onto object. Then print the program's details, including the program level.

1.2 Class `Land` has the following attributes and methods:

Attributes:
- id
- owner name
- house type
- area

Methods:
a) Constructor
b) Normal constructor
c) Copy Constructor
d) Mutator/Setter
e) Accessor/Getter
f) Processor – Calculate tax

The tax on this type of land depends on its area, and the type of house built on the land as shown in the following table:

House Type	**Description**	**Tax rate (RM/m^3)**
T	Terrace	10
S	Semi-Detached	15
B	Bungalow	20
C	Condominium	30

Details of land

g) Printer to return object information.

Write a program to read the id, owner name, house type, area of land and store them onto object. Then, print the details of land, including the tax price.

1.3 The class `Cinema` has the following instance data:

- Cinema code – example: `LFS1, LFS2, LFS3`
- Movie title – example: `Maleficient, X-Men`
- Price
- Mode payment – example: `credit card, cash`
- Membership – Example: `true` if member otherwise `false`

a) Write a complete class called `Cinema` with the following methods:

i. Default/Normal and Copy constructors

ii. Mutator/Setter method for each data member

iii. Accessor/Retriever/Getter methods for all data members

iv. `toString()` method to return the objects' information

v. The method `discount()`: a processor method to return discount rates based on the mode of payment and membership of every customer. The range is given based on the following table:

<table>
<tr><th>Mode of payment</th><th>Membership</th><th>Discount Rate</th></tr>
<tr><td rowspan="2">Credit card</td><td>true</td><td>5%</td></tr>
<tr><td>false</td><td>3%</td></tr>
<tr><td rowspan="2">cash</td><td>true</td><td>10%</td></tr>
<tr><td>false</td><td>0%</td></tr>
</table>

b) Write an application class called `CinemaApp` to perform the following:

i. Declare an object of Cinema and store the data

ii. Display the details of Cinema.

1.4 The class `Transport` has the following data members:

- Registration number – example: CAA123, PUM 456
- Brand – Toyota, Honda, Proton
- Purchase price – Example: RM 120,000

a) Write a complete class called `Transport` with the following methods:

i. Default/Normal/Copy constructors

ii. Mutator/Setter method for all data members

iii. Accessor/Retriever/Getter methods for all data members

iv. `toString()` method to return the objects' information

v. `TransportSelangor()` : The method will determine whether the transport was registered in Selangor or not and return the true/false value. This is determined by the transport registration number (e.g: BBJ 383, if it starts with code 'B', it means that transport was registered in Selangor and true value will be returned, otherwise false.

vi. `DiscountPrice()`: To calculate and return the price for transport. For each transport the price is based on the discount rate. The discount rate is given through the parameter.

b) Write an application class called `TransportApp` to perform the following:

i. Declare an object of Transport and store the data

ii. Display the details of Transport.